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Probabilistic Combinatorics SS24

Sheet 10

Due: 24. June at 19:00

(1) Formulate and proof analogs of the FKG Inequality for the cases, where

- f is monotonically decreasing and g monotonically increasing.
- f and g are both monotonically decreasing.

(2) Follow from the FKG Inequality, that for real numbers $0 \leq a_1 \leq ... \leq a_n$ and $0 \leq b_1 \leq ... \leq b_n$ it holds, that

$$\langle a, b\rangle \geq \frac{\langle a, \mathbf{1}\rangle \cdot \langle \mathbf{1}, b\rangle}{n},$$

where $\mathbf{1} \in \mathbb{R}^n$ is the constant 1-vector and a and b are the vectors with ordered entries a_i resp. b_i. $\langle \cdot, \cdot \rangle$ is the standard dot product.

(3) Let $p \in [0, 1]$. From the set $[n]$ there will be chosen a subset A randomly, by choosing independently with probability p for each $i \in [n]$ whether i is taken into A.

From A there will be constructed $B \subseteq A$, by removing every i with $\{i, i+1\} \subseteq A$.

Determine the expected value of the cardinality of B and determine p such that

$$\lim_{n\to\infty} |\mathbb{E}[\#B] - \frac{n}{4}| = 0.$$

(4) There are new rules for the exams of the hessian Abitur: There are exams in n subjects. Every student will take the exams in all of the n subjects. Each student may choose in every subject whether they want to take the exam in English or in German.

In 2024 there were m students. For every pair of subjects there was at least one student, who took the exams in those two subjects in different languages.

Proof: In 2024 there were no more than 2^m subjects.